MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1603.3
STANDARD	
Time Transfer Metadata	**18 October 2022**

1 Scope

This standard defines the metadata constructs for supporting Time Transfer, the process of communicating accurate time from a reference clock (e.g., GPS device) to a receptor clock (e.g., camera). This standard builds upon the MISP Time System's timestamp definitions in MISB ST 0603 [1].

This standard defines two metadata constructs: **Time Transfer Local Set** and **Nano Time Transfer Pack**. The **Time Transfer Local Set** provides metadata to validate and correct timestamps, along with metadata to aid in converting a MISP timestamp to Universal Coordinated Time (UTC). Please refer to the Motion Imagery Handbook for more information on conversions between MISP Time and UTC.

The **Nano Time Transfer Pack**, is a KLV pack consisting of a Nano Precision Time Stamp, as defined in MISB ST 0603, and the Time Transfer Local Set.

2 References

[1] MISB ST 0603.5 MISP Time System and Timestamps, Oct 2017.

[2] MISB MISP-2023.1: Motion Imagery Handbook, Oct 2022.

[3] MISB ST 0601.18 UAS Datalink Local Set, Oct 2022.

[4] MISB ST 0807.26 MISB KLV Metadata Registry, Jun 2022.

[5] MISB ST 0107.4 KLV Metadata in Motion Imagery, Feb 2019.

[6] Assistant Secretary of Defense for Command, Control, Communications and Intelligence Global Positioning Standard Positioning Service Performance Standard 4th Ed, Sept 2008.

[7] IEEE Standard 1588-2002.1 IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems, 2002.

[8] IEEE Standard 1588-2008.2 IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems, 2008.

[9] IETF RFC 1305 Ver. 3 .3 Network Time Protocol, 1992.

[10] IETF RFC 5905 Ver. 4 Network Time Protocol, Jun 2010.

[11] IRIG Standard 200-04 Telecommunications and Timing Group/Range Commanders Council, Sep 2004.

3 Acronyms and Definitions

CRC	Cyclic Redundancy Check
GPS	Global Positioning System
IERS	International Earth Rotation and Reference Systems Service
IRIG	Inter-range Instrumentation Group
KLV	Key Length Value
MISB	Motion Imagery Standards Board
MISP	Motion Imagery Standards Profile
NTP	Network Time Protocol
PPS	Pulse Per Second
PTP	Precision Time Protocol
SI	International System (of units)
TAI	International Atomic Time
UTC	Universal Coordinated Time

4 Revision History

Revision	Date	Summary of Changes
ST 1603.3	10/18/2022	• Added Requirement -13 • Updated UTC time conversion commentary to use the Motion Imagery Handbook algorithm • Changed document title from “Time Transfer Pack” to “Time Transfer Metadata” • Changed “element” to “Item” for consistency with other MISB usage • Non-material edits to improve readability • Removed Time Transfer Local Set item’s KLV keys • Updated References

5 Introduction

The MISP Time System, defined in the Motion Imagery Handbook [2] and its requirements specified in MISB ST 0603, is a “Level 3” time system exemplifying the following characteristics:

1. Total Order of data
2. Relative Differencing
3. Absolute Time reference

Locally, the MISP Time System derives its count using an internal clock, which references an external clock source, such as a Global Positioning System (GPS) receiver, Inter-range Instrumentation Group (IRIG) system, or a Precision Time Protocol (PTP) device. A group of highly accurate atomic clocks form the foundation of time distribution, where a time signal –

called the reference clock, distributes to one or more local clocks – called a receptor clock. Time Transfer is the process of communicating time from the reference clock to a receptor clock.

Figure 1 illustrates a time transfer system where the goal is to transfer the time established by a referenced atomic clock to a receptor clock used by a sensor.

There are two time-transfer operations occurring in Figure 1: the first is the atomic clock to a receiver; the second is the receiver to a sensor. This standard provides metadata to qualify the time transfer process between a receiver (Reference) and a sensor (Receptor). This standard does not address the time transfer process between an atomic clock and a receiver, which being integral to the receiver electronics is difficult to ascertain. In Figure 1, with the Receiver synchronized to the Atomic Clock, the Receiver now becomes the Reference Signal source for the Sensor's local clock. The Receiver produces both a time value, and a Reference Signal. The Reference Signal is a highly accurate pulse aligned to the exact time when an SI second (or fraction thereof) occurs.

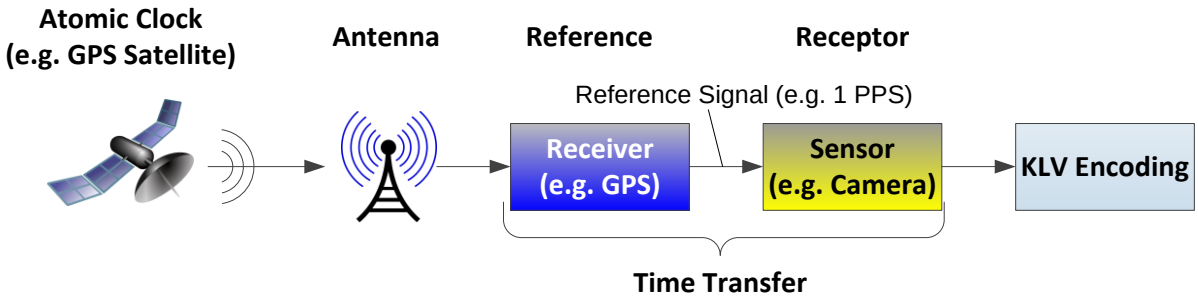


Figure 1: Time Transfer System

In practical application, continuous Time Transfer is not 100% guaranteed. For instance, a receptor clock may temporarily lose its ability to report time at a Level 3, whereby it then produces Level 0 timestamps instead. By tracking the Time Transfer process, and reporting this information as metadata, post processing can correct the reported time to Level 3. Furthermore, using a corrective set of rules the reported time during Time Transfer loss can guarantee at least a Level 1 timestamp. To perform correction, and optionally guarantee a minimum of Level 1 performance, this standard defines the Time Transfer Local Set. Consult the Motion Imagery Handbook for details on the method to adjust a receptor clock.

To achieve Level 3, a MISP Time System is a count of SI seconds represented with a specific resolution (e.g., nanoseconds) beginning at the MISP Epoch without adjustment for leap seconds. Since the MISP Time System's times are not equal to UTC, when reporting time from a UTC source, receivers must convert the time to include the leap second adjustment. See the Motion Imagery Handbook for information on converting to/from UTC to a MISP Time System time. The International Earth Rotation and Reference Systems Service (IERS) can update leap second information over any 6-month period, so the Time Transfer Local Set includes leap second information to track the current adjustment value. Some receivers (e.g., GPS) provide leap second information along with their timing signals.

6 Time Transfer Local Set

The purpose of the Time Transfer Local Set is to qualify the relationship between a receptor clock and its reference clock with status information. The reference clock is a highly accurate reference signal, which a receptor can synchronize its clock. When there is a disruption or loss of the reference signal, the receptor clock "free-wheels" (i.e., drifts) slowly diverging from the reference clock. Upon reacquiring the reference signal, any temporal difference between the two clocks requires a correction. The Time Transfer Local Set supports two methods for correcting the difference: Jam Correction and Slew Correction.

Jam Correction forces an abrupt change to a receptor's clock report time to the latest reported time of the reference clock. If the receptor clock is free-wheeling faster than the reference clock, the reported time forces a negative jump (i.e., backward) in time creating a negative timing discontinuity. Such discontinuities change the system to Level 0 and complicates downstream processing.

Slew Correction slows down the receptor clock allowing the reference clock to "catch up". During this period of slewing, systems provide the difference between the reported time (receptor clock) and the reference time (reference clock) as metadata; this metadata enables an application to correct time instantaneously, if needed. Proper initialization in Slew Correction ensures monotonicity. When used, Slew Correction maintains a minimum of a Level 1 Timing System, insuring Total Ordering. Section 8 provides the rules for Slew Correction.

The Time Transfer Local Set applies to the timestamp defined in the metadata set it appears; to generalize, this timestamp is the *parent time*. For example, within a MISB ST 0601 [3] UAS Datalink Local Set, the Time Transfer Local Set applies to the Precision Time Stamp value (Tag 2), which in this case is the parent time. Every instance of a parent timestamp within a metadata set is a *time report*. This standard requires certain Time Transfer metadata items at a minimum periodicity, while systems may include other items under certain conditions, on a less-frequent basis.

The Time Transfer Local Set always embeds within another metadata set or group. For this reason, the Time Transfer Local Set does not include a CRC value, because the metadata set or group containing the parent time should already provide CRC validation.

MISB ST 0807 [4] registers the Time Transfer Local Set 16-Byte Universal Label as:

06.0E.2B.34.02.0B.01.01.0E.01.03.02.02.00.00.00 (CRC 43211)

6.1 KLV Metadata in Motion Imagery

MISB ST 0107 [5] provides a set of baseline requirements for applying KLV metadata in Motion Imagery.

Requirement	
ST 1603.3-13	All Time Transfer Local Set metadata shall be in accordance with MISB ST 0107.

6.2 Local Set Items

Table 1 lists the Time Transfer Local Set items with the following column designations:

- **Tag** is the Local Set tag number for the item in the Time Transfer Local Set.
- **Item Name** is the name of the item.
- **Section** is the section number in this document providing details for the item.
- **Type** indicates the data type of the item's value.
- **Units** provides the unit of specification for the value.
- **Usage** is the reporting frequency (i.e., once per 30 seconds), or the reporting conditions.

Table 1: Time Transfer Local Set

Time Transfer Local Set					
Tag	**Item Name**	**Section**	**Type**	**Units**	**Usage**
1	Document Version	6.3.1	uint	N/A	Use at least once per 30 Seconds
2	UTC Leap Second Offset	6.3.2	int	Seconds	Use at least once per 30 Seconds
3	Time Transfer Parameters	6.3.3	uint	N/A	Use at least once per 30 Seconds
4	Synchronization Pulse Frequency	6.3.4	float	Hz	Use at least once per 30 Seconds
5	Unlock Time	6.3.5	uint	See Section 6.3.5	Use only when lost lock
6	Last Synchronization Difference	6.3.6	uint	See Section 6.3.6	Use only when providing Slew Correction
7	Drift Rate	6.3.7	float	μs/s	Use at least once per 30 Seconds
8	Signal Source Delay	6.3.8	uint	ns	Use at least once per 30 Seconds
9	Receptor Clock Uncertainty	6.3.9	uint	See Section 6.3.9	Use at least once per 30 Seconds

6.3 Item Details

6.3.1 Document Version

The Document Version identifies the version of MISB ST 1603 for the Time Transfer Local Set. The value is set to the minor version of the document; for example, ST 1603.1 would have a value of 1. The Document Version is present in the stream at least once every thirty seconds.

Requirement	
ST 1603-01	The Document Version shall be included in the Time Transfer Local Set at least once every 30 seconds.

6.3.2 UTC Leap Second Offset

The UTC Leap Second Offset is an integer number of SI seconds needed to convert a timestamp to Universal Coordinated Time (UTC). The leap seconds offset accounts for leap seconds since the MISP Epoch as defined in the Motion Imagery Handbook. For GPS systems, the leap second

difference between the MISP Epoch and the GPS Epoch is 19-8=11. When GPS reports a leap second offset (relative to the GPS Epoch) add the value of 11 to make it relative to MISP Epoch.

This standard requires the inclusion of the UTC Leap Second Offset in a Motion Imagery stream along with time reports; however, this value does not need to be present with every time report. The UTC Leap Second Offset value usually changes at most once a year. During regular operation, systems may deliver the UTC Leap Second Offset in the Time Transfer Local Set as infrequently as once every 30 seconds.

Some systems, such as GPS, announce months in advance when the next leap second adjustment will occur. When a leap second change is going to occur within the next 30 seconds, include the UTC Leap Second Offset with every reported time. During and after the leap second change, include the UTC Leap Second Offset with every reported time for 30 seconds after the change. After the update, the UTC Leap Second Offset reports at its nominal rate, but no more than 30 seconds apart.

Requirement	
ST 1603-02	The UTC Leap Second Offset shall be included in the Time Transfer Local Set at least once every 30 seconds.

6.3.3 Time Transfer Parameters

The Time Transfer Parameters item is a binary value composed of three sub-items: The Reference Source Status, Correction Method, and Time Transfer Method.

Although it is not necessary to report the Time Transfer Parameters value for every time report, it must appear in a Motion Imagery stream at least once every 30 seconds.

Requirement	
ST 1603-03	The Time Transfer Parameters item shall be included in the Time Transfer Local Set at least once every 30 seconds.

Figure 2 illustrates the binary format for the Time Transfer Parameters, which is one-byte long (this could become larger in future versions of this standard). The least two significant bits (i.e., bit 0 and bit 1) indicate the Reference Source (see Section 6.3.3.1); the next two significant bits (i.e., bit 2 and bit 3) indicate the Correction Method (see Section 6.3.3.2); the next four significant bits (bit 4, 5, 6 and 7) indicate the Time Transfer Method (see Section 6.3.3.3).

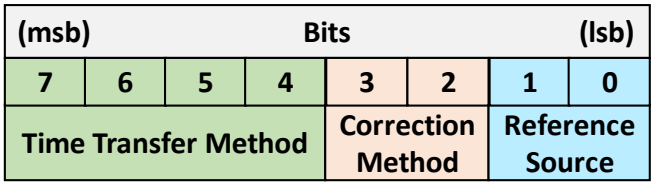


Figure 2: Time Transfer Parameters

When future versions add additional bytes, the additional bytes will be the more significant bytes, where the byte shown in Figure 2 will become the Least Significant Byte (LSB).

6.3.3.1 Reference Source

The Reference Source (bits 0, 1) sub-item indicates if the reference clock is synchronized with an atomic source (i.e., atomic clock), or not. For example, in normal operations, a GPS synchronizes to satellites, and effectively, an atomic clock. However, if the antenna breaks, the GPS can no longer synchronize to the atomic clock. When the reference clock is synchronized to an atomic source (e.g., TAI or GPS), the Reference Source value is set to two (2); otherwise, it is set to one (1). When the state of the clock is unknown, use a value of zero (0). Table 2 summarizes the allowed values for the Reference Source.

Table 2: Allowed values for Reference Source

Value	Meaning
0	Reference Source status is unknown
1	Reference Source is not synchronous to an atomic source
2	Reference Source is synchronous to an atomic source
3	Reserved for future use

6.3.3.2 Correction Method

The Correction Method (bits 2, 3) sub-item indicates Unknown, Jam Correction, or Slew Correction (see Section 8 for details on Correction Method application). Table 3 lists the valid values for the Correction Method. Unknown indicates either no correction, or a method other than Jam/Slew Correction.

Table 3: Allowed values for Correction Method

Value	Meaning
0	Unknown
1	Jam Correction
2	Slew Correction
3	Reserved for future use

6.3.3.3 Time Transfer Method

The Time Transfer Method (bits 4 - 7) sub-item indicates the system used to transfer time from a reference clock to a receptor clock. Table 4 lists the valid values for the Time Transfer Method. A value of zero (0) alerts users about the uncertainty in the creation of the reported time.

Table 4: Allowed values for Time Transfer Methods

Value	Meaning
0	Unknown Time Transfer Method
1	Global Positioning System (GPS) PPS [6]
2	Precision Time Protocol (PTP) - Version 1 [7]
3	Precision Time Protocol (PTP) – Version 2 [8]
4	Network Time Protocol (NTP) [9]
5	Network Time Protocol (NTP) [10]
6	Inter-range Instrumentation Group (IRIG-A) [11]
7	Inter-range Instrumentation Group (IRIG-B) [11]
8-15	Reserved for future use

6.3.4 Synchronization Pulse Frequency

The Synchronization Pulse Frequency item specifies the periodicity of the pulse transmitted by a reference time source. The unit of measure is Pulse per [SI] Second (PPS) or Hertz (Hz). This standard does not require reporting the Synchronization Pulse Frequency; its default value is one (1.0 PPS). When providing the Synchronization Pulse Frequency value, it must appear in the stream at least once every 30 seconds.

Requirement	
ST 1603-04	While the Synchronized Pulse Frequency is not equal to one (1.0) Pulse per [SI] Second (PPS), the Synchronized Pulse Frequency shall be included in the Time Transfer Local Set at least once every 30 seconds.

6.3.5 Unlock Time

The Unlock Time is the duration of time since the receptor clock and the reference clock (see the Motion Imagery Handbook for details) were last in lock. When unlocked, provide the Unlock Time value for every time report. The default value is zero (0). With a locked reference and receptor clock, this value is zero (0), and therefore, there is no need to report the value. A non-zero value indicates the duration the receptor and reference clocks have not been locked. This replaces the locked/unlocked bit used in other standards. The units for this measurement are the same as the units of the parent time.

Requirement	
ST 1603-05	While the Unlock Time is greater than zero (0), Unlock Time shall be included in the Time Transfer Local Set.

6.3.6 Last Synchronization Difference

When performing Slew Correction, the Last Synchronization Difference item specifies either the current difference in time between a receptor clock and its reference clock during slewing, or the

last known greater-than-zero difference during slewing (see the Motion Imagery Handbook for a description). When using Slew Correction, and the Slew Correction value is greater than zero (0), report this value with every time report. When the value is zero, do not include the value in the Local Set, i.e., the default value is zero (0). The units for this measurement are the same as the units of the parent time.

Requirement	
ST 1603-06	Where using Slew Correction, when the Last Synchronization Difference is greater than zero, the Last Synchronization Difference shall be included in the Time Transfer Local Set.

6.3.7 Drift Rate

The Drift Rate item specifies the known maximum drift rate of the receptor clock when free-wheeling. When providing the Drift Rate, it must appear in the stream at least once every 30 seconds. Omitting this value from the stream, means the drift rate is unknown, i.e., there is not a default value. This value can be a positive or negative floating-point value, and the units are in microseconds per second (μs/s).

When including the Drift Rate in the Time Transfer Local Set, the reported time's uncertainty is known in real time when losing the reference source.

Requirement	
ST 1603-07	Where the Drift Rate is included, the Drift Rate shall be included in the Time Transfer Local Set at least once every 30 seconds.

6.3.8 Signal Source Delay

The Signal Source Delay item is the estimated or measured latency of the time signal from the reference clock to the receptor clock. For example, on-board a ship a GPS antenna could be 200 meters from a receiver, where the signal incurs a non-zero latency. When providing the Signal Source Delay value, it must appear in the stream at least once every 30 seconds. The default Signal Source Delay is zero (0). The units for this measurement are nanoseconds.

Requirement	
ST 1603-08	Where the Signal Source Delay is greater than zero (0), it shall be included in the Time Transfer Local Set at least once every 30 seconds.

6.3.9 Receptor Clock Uncertainty

The Receptor Clock Uncertainty item provides the estimated error of the receptor clock (i.e., standard deviation of the measured time, see [2]). When providing the Receptor Clock Uncertainty, it must appear in the stream at least once every 30 seconds. Omitting Receptor Clock Uncertainty in the stream, means the Receptor Clock Uncertainty is unknown. The units for this measurement are the same as the units of the parent time.

Requirement	
ST 1603-09	Where the Receptor Clock Uncertainty is included, it shall be included in the Time Transfer Local Set at least once every 30 seconds.

7 Nano Time Transfer Pack

The Nano Time Transfer Pack is a KLV construct composed of a two-element truncation pack. The first element is the Nano Precision Time Stamp as defined in MISB ST 0603, which is a 64-bit unsigned integer representing time measured from the MISP Epoch in nanoseconds. The second element is the value portion of the Time Transfer Local Set. A truncation pack saves bandwidth and co-joins the time transfer metadata to be directly associated with its parent time within the same KLV group. Figure 3 illustrates the two-value pack layout.

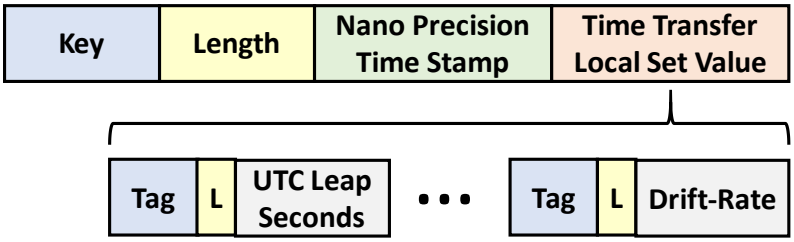


Figure 3: Illustration of Nano Time Transfer Pack

The KLV Key for the Nano Time Transfer Pack registered in MISB ST 0807 is:

06.0E.2B.34.02.05.01.01.0E.01.03.02.09.00.00.00 (CRC 58798)

The Nano Precision Time Stamp represents a sampled and quantized time value of the MISP Time System.

8 Clock Corrections

As discussed in Section 6, there are two methods for correcting a clock value when reacquiring the lock: Jam Correction and Slew Correction. Jam Correction allows a receptors clock value to change negatively, so it matches the receiver's clock. Slew Correction prevents the time value from changing negatively by slowing incrementing the receptors clock while the receiver's clock catches up.

Implementing Jam Correction does not require any special metadata in the Time Transfer Local Set.

Implementing Slew Correction requires the use of the Last Synchronization Difference (Section 6.3.6), and some additional requirements on the receptor clock. With Slew Correction, the time value must always increase, so during power-up or initialization (before the first reference lock) the clock value must provide a time that is earlier than the current time. Starting devices using a timestamp value of zero ensures a positive jump when systems are initializing prior to synchronization to a reference clock.

Requirement(s)	
ST 1603-11	Where using Slew Correction, time values shall be monotonically increasing.
ST 1603-12	Where using Slew Correction, the time values before the first reference clock lock shall start at zero (0) and increment at the receptor clocks rate.

9 Retired Requirements

Requirement	
ST 1603-10 (Retired)	The 64-bit Nano-Second Counter shall represent a value of the MISP Time System truncated to the nearest nanosecond. [Note: requirement moved to MISB ST 0603]